

Day 22 - A/B Experimentation & Data-Subject Rights

PlaceMux Phase 2 | Data Analyst | Task 22 | Week 6 | Phase 3

Item	Verified result
Experiment ID	exp_offer_nudge_v1
Subjects	6,000
Eligible	4,874
Exposed	4,796
Control / Treatment	2,385 / 2,411
Primary metric	7-day application submission
Treatment - Control	+0.57 pp
P-value	0.6227
Validation	PASS

This report documents the demo-ready experimentation framework, its data-subject controls, resilience checks, statistical comparison and actual dashboard views. It intentionally excludes the written submission answer.

1. Task Scope & Implementation

The Task 22 brief asks to stand up the experimentation framework and considers the task complete when the A/B framework is ready. The implementation therefore focuses on a governed, auditable experiment contract rather than pretending that synthetic performance is production PlaceMux performance.

- **Experiment contract:** hypothesis, primary metric, denominator, analysis window, guardrails and decision rule.
- **Assignment:** deterministic 50/50 allocation using a stable subject identifier and versioned assignment rule.
- **Exposure:** assignment is separated from actual exposure, with exposure logged explicitly.
- **Rights:** consent, withdrawal and deletion states are checked before exposure and analysis.
- **Resilience:** duplicate, assignment, lifecycle, variant and allocation-balance checks are machine validated.

2. Experiment Definition

Field	Definition
Hypothesis	Treatment nudge increases 7-day application submission.
Primary metric	application_submitted_7d
Assignment	Deterministic 50/50 split on stable subject ID hash
Eligibility	Consent granted; no deletion request; no experiment withdrawal; eligibility score threshold
Primary analysis	Exposed eligible subjects with stable assignment
Guardrails	Exposure, application start, interview completion, offer rate, and data-subject rights integrity
Decision	Ship only when the pre-registered primary rule is met and guardrails pass

3. Metric & Tracking Framework

The repository separates measurement definitions from event instrumentation. The metric contract fixes denominators and first-occurrence rules; the tracking plan records assignment and exposure as distinct lifecycle events and adds explicit rights events for consent withdrawal and deletion requests.

Metric	Formula / rule
Eligible subjects	COUNT(DISTINCT subject_id) meeting consent and experiment eligibility
Exposure rate	exposed / assigned
Primary conversion	primary successes / pre-registered analysis population
Absolute lift	treatment conversion - control conversion
Relative lift	absolute lift / control conversion
95% CI	Treatment-control interval for binary primary outcome
Rights breach count	Exposure or analysis after a blocking rights state should have applied

Core tracked events: experiment_eligible, experiment_assigned, experiment_exposed, application_started, application_submitted, interview_completed, offer_received, consent_withdrawn, deletion_requested.

Logical model: Experiment -> Assignment -> Exposure -> Outcome Events, governed by a Subject Consent / Rights State. The derived analysis view retains first valid assignment, first valid exposure, primary outcome, secondary outcomes and exclusion reason.

4. Dry-Run Results

The dry run uses a synthetic population of 6,000 subjects. Results below demonstrate calculation mechanics only.

Variant	n	Primary successes	Primary rate	95% CI / lift
Control	2,385	478	20.04%	-
Treatment	2,411	497	20.61%	-1.71 to +2.85 pp

Observed treatment-control difference: **+0.57 percentage points** (+2.9% relative lift), with p-value **0.6227**. The 95% interval is **-1.71 to +2.85 percentage points**, so the dry-run result does not justify a statistically confident production ship decision by itself.

This interpretation is intentionally conservative: the framework separates mechanical readiness from a business decision. Production experimentation should use a pre-registered analysis, approved thresholds, appropriate power/sample-size planning, and governed statistical methodology.

5. Actual Dashboard Evidence - Overview & Design

PlaceMux A/B Experimentation & Data-Subject Rights Framework

Synthetic experimentation dry run · exp_offer_nudge_v1 · stable assignment · consent-aware exclusion · resilience controls

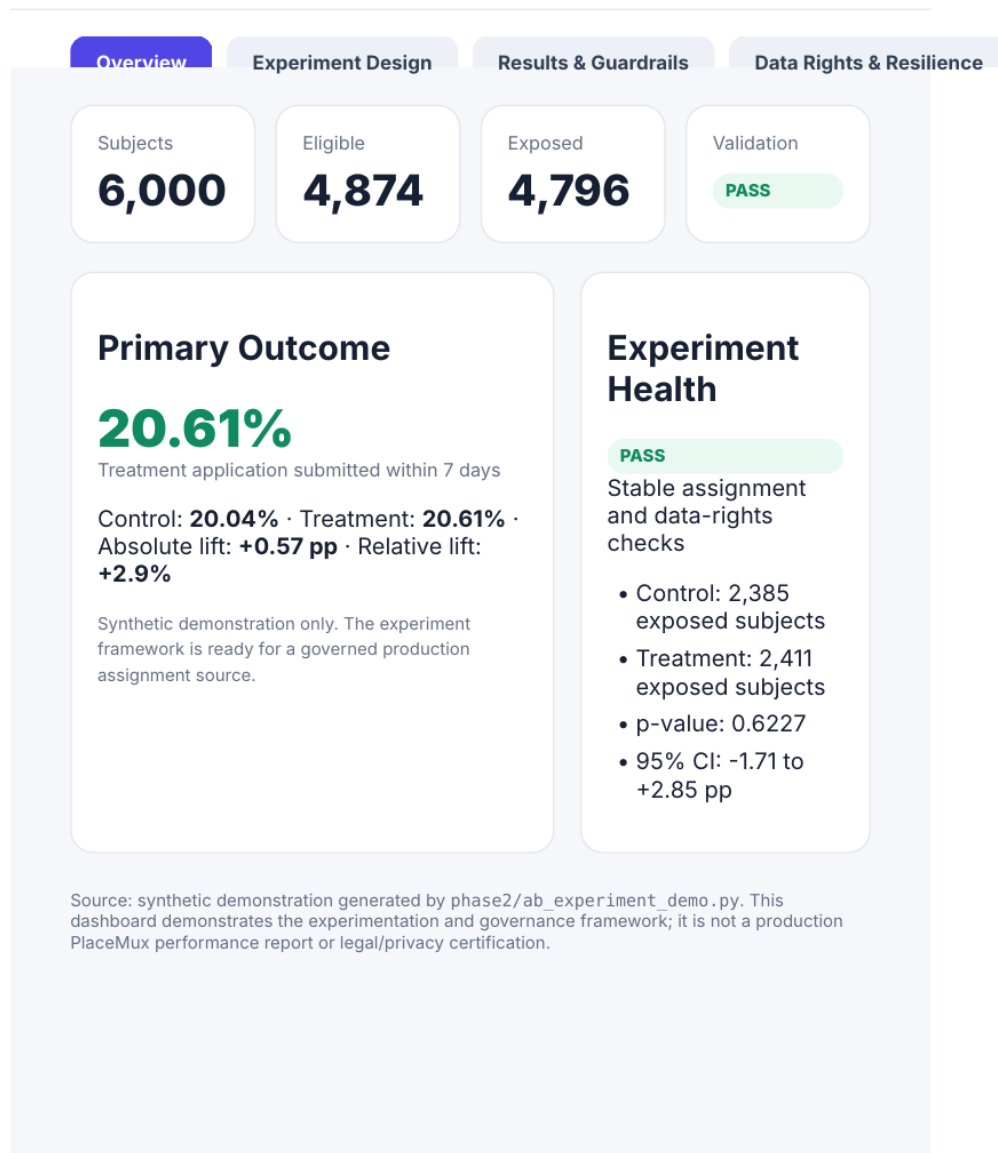


Figure 1. Actual dashboard overview rendered from the completed HTML dashboard.

PlaceMux A/B Experimentation & Data-Subject Rights Framework

Synthetic experimentation dry run · exp_offer_nudge_v1 · stable assignment · consent-aware exclusion · resilience controls



Figure 2. Actual experiment design view rendered from the completed HTML dashboard.

6. Actual Dashboard Evidence - Results & Data Rights

PlaceMux A/B Experimentation & Data-Subject Rights Framework

Synthetic experimentation dry run · exp_offer_nudge_v1 · stable assignment · consent-aware exclusion · resilience controls

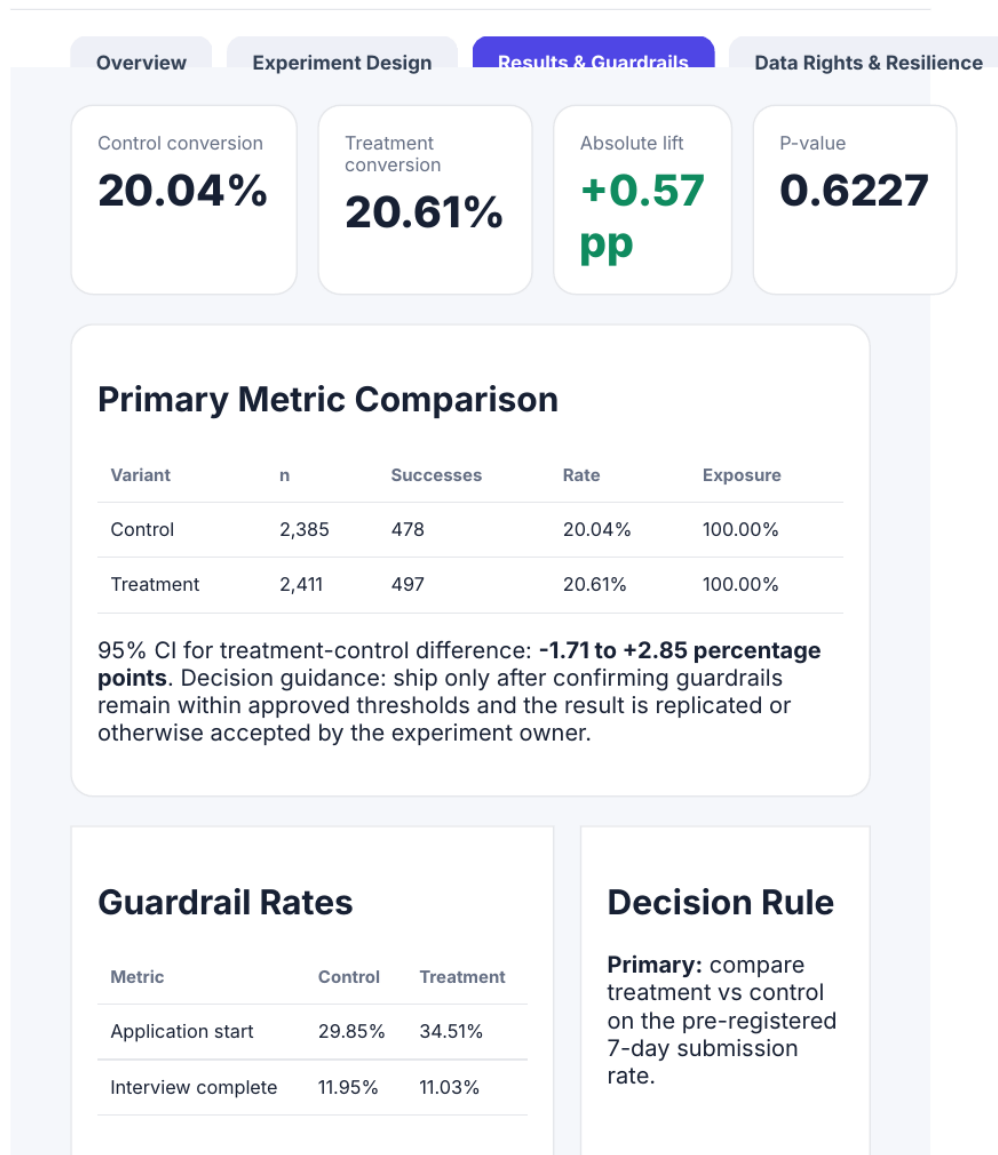


Figure 3. Actual A/B results and guardrail view rendered from the completed HTML dashboard.

PlaceMux A/B Experimentation & Data-Subject Rights Framework

Synthetic experimentation dry run · exp_offer_nudge_v1 · stable assignment · consent-aware exclusion · resilience controls

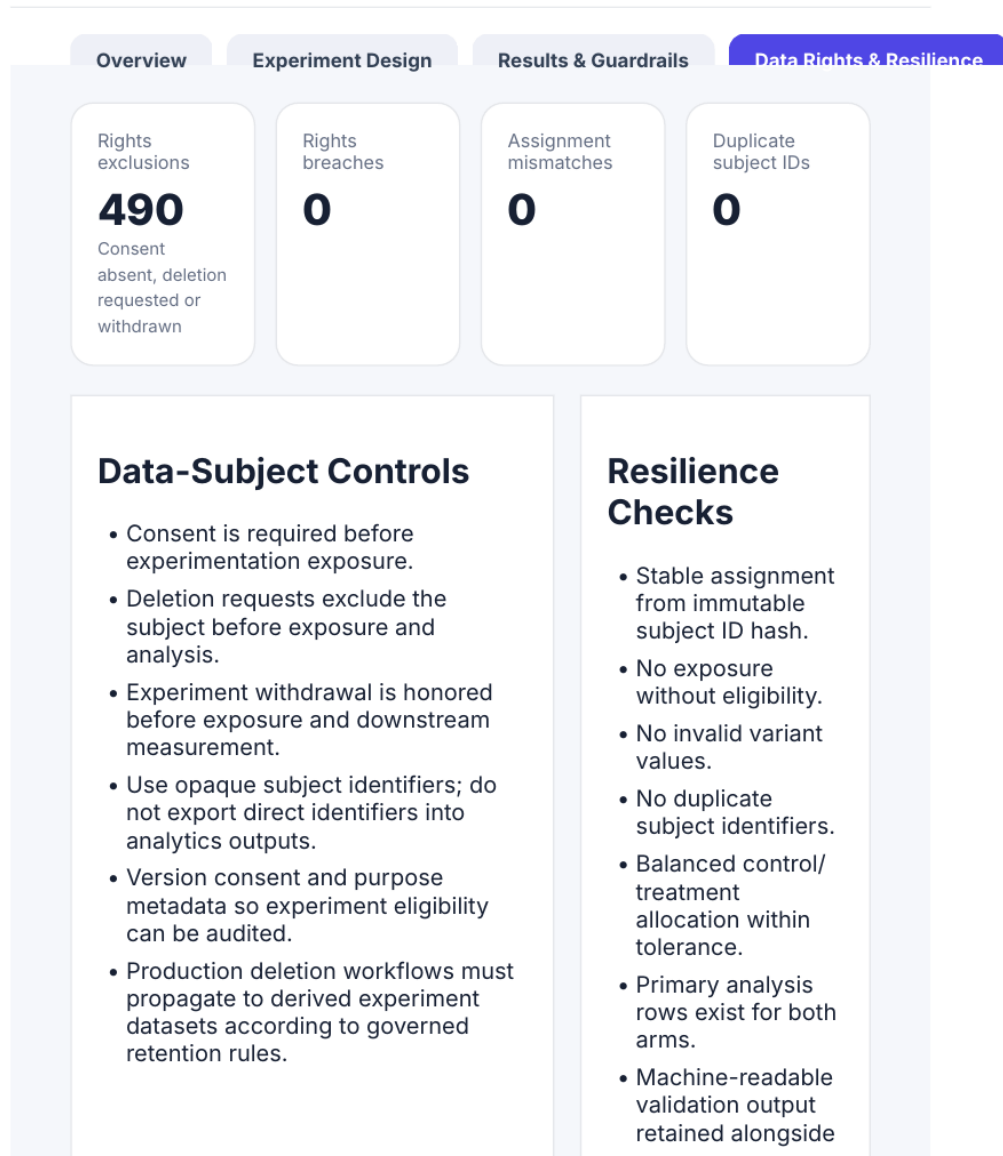


Figure 4. Actual data-rights and resilience view rendered from the completed HTML dashboard.

7. Validation & Reproducibility

Validation check	Result
Duplicate subject IDs	0
Assignment mismatches	0
Exposed without eligibility	0
Data-subject rights breaches	0
Invalid variant values	0
Missing analysis rows	0
Arm-size imbalance	26
P-value present	0
Overall status	PASS

Repository artifact chain:

- `phase2/ab_experiment_metrics.md` - metric definitions and denominator rules.
- `phase2/ab_experiment_tracking_plan.md` - event taxonomy, assignment/exposure contract and rights controls.
- `phase2/ab_experiment_model.md` - logical data model and derived analysis view.
- `phase2/ab_experiment_demo.py` - deterministic synthetic data, calculations, validation and dashboard generation.
- `phase2/ab_experiment_events_demo.csv` - synthetic subject-level experiment source.
- `phase2/ab_experiment_results.csv` - arm-level results.
- `phase2/ab_experiment_validation.json` - machine-readable validation evidence.
- `phase2/ab_experiment_dashboard.html` - self-contained local demo dashboard.

Exact reproduction command: `python phase2/ab_experiment_demo.py`

Scope note: This is a synthetic experimentation dry run. It is not a production PlaceMux performance report, statistical certification, legal/privacy certification, or security certification.